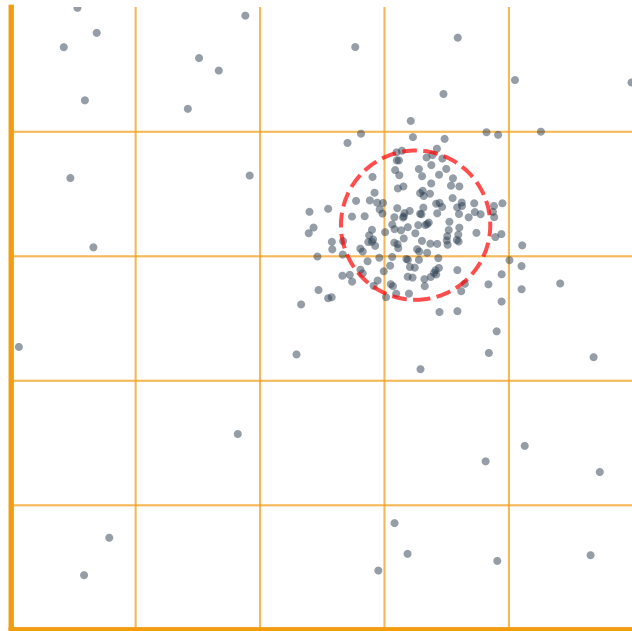
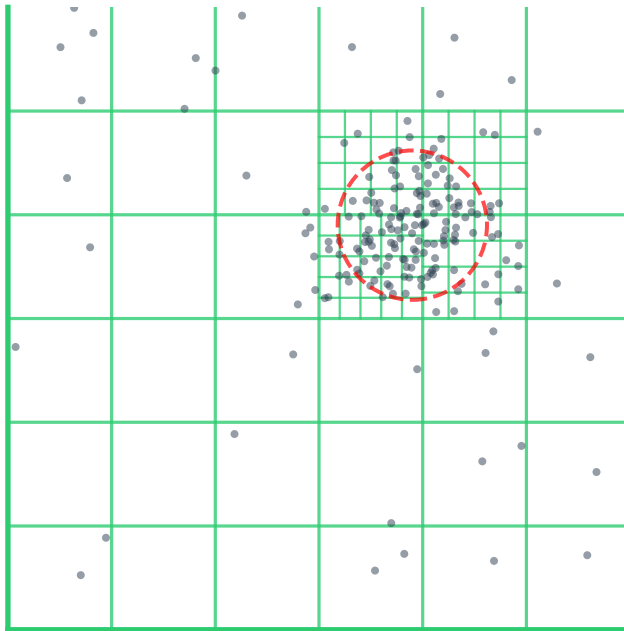


Spatial Index Subdivision Comparison (N=200 clustered points)

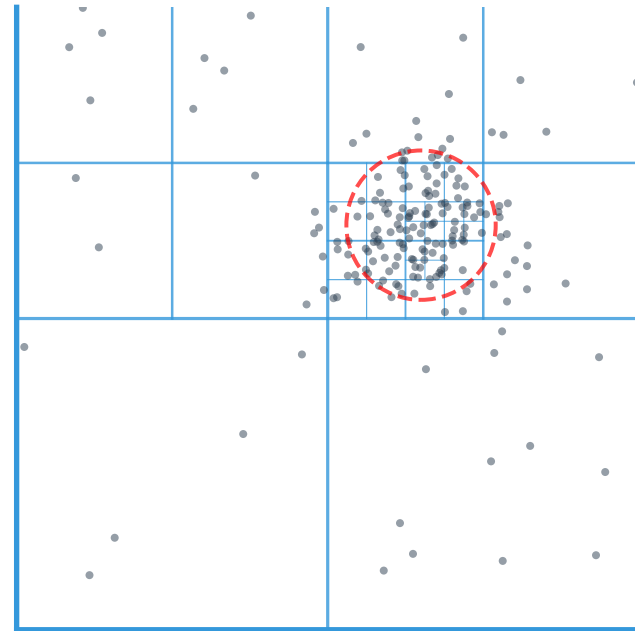
Uniform Grid
 $g \times g$ fixed ($g=5$)



RDT
 $g(n) = \text{floor}(\log(n+1)^\alpha)$



Quadtree
4-way split at threshold



Red circle marks the dense cluster. RDT uses log-based adaptive subdivision; Quadtree always splits 4 ways; Uniform Grid ignores density entirely.